Arthur Lu

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EDUCATION

University of Southern California — Los Angeles, CA
Master of Science in Computer Science. — May 2028 (Expected)
University of Nottingham Ningbo China — Ningbo, China
B.S. (Hons), Computer Science with Artificial Intelligence. — Jul 2026
Second-Class Honours (Division One).

EXPERIENCE

Undergraduate Research Assistant — Jun 2025 - Aug 2025
University of Nottingham Ningbo China — Ningbo, China

- Owned HoloLens 2 integration within a **6-person team**, connecting a Python/OpenPose pose-processing pipeline with C#/Unity feedback components to support interactive assessment of 6 rehabilitation movements.
- Implemented geometry-based estimation against an existing 3D human model to recover **occluded joints** and calculate ankle and toe angles; evaluated the pipeline on video clips, including foot flexion and extension movements.

Undergraduate Research Assistant — Jun 2024 - Aug 2024
University of Nottingham Ningbo China — Ningbo, China

- Led a **7-person team** to deliver a Raspberry Pi/Arduino security prototype: defined module interfaces, coordinated integration of entry detection, alarms, and camera capture, and implemented control code for its hardware components.

Software Development Intern — Jan 2024 - Feb 2024
Shenzhen Topsky Software — Shenzhen, China

- Owned 3 native **WeChat Mini Program** pages for vaccine details, purchase, and appointment booking; passed QA acceptance and shipped to a **nationwide healthcare** booking platform with 4,000 partners, 600+ cities, 67M+ cumulative members.
- Integrated 3+ backend APIs to feed the product-selection popup with images and prices and link it to appointment booking; corrected object-copying defects that produced **duplicate product entries**, restoring consistent selection behavior before release.

PROJECTS

RunSleuth | PyTorch Training Diagnostics & Repair Verification — Sep 2026 - Present

- Independently built a tool that automatically **diagnoses and repairs silent bugs** in PyTorch training; reproduced and repaired 4 **silent training faults** on ResNet18 with 20K Camelyon17-WILDS images, each verified by bounded retraining across 2 seeds.
- Exposed **LLM overrides of correct evidence** in a 12-case optimizer challenge comparing rules, an LLM, and a **Gemini 3.8 Flash** tool-calling agent with telemetry-validated citations; also designed a failure-signature library that diagnosed 18/18 runs correctly on a self-built benchmark with a healthy reference, with 0 false positives on healthy runs.

CMCD | Cross-Modality MRI Translation — Jun 2025 - May 2026

- Independently engineered a PyTorch **MRI-translation pipeline** spanning NIfTI preprocessing, conditional diffusion training, checkpoint recovery, and inference; processed **20K+ BraTS 2020 slices** into standardized 256x256 inputs for T1-to-T2 synthesis.
- Combined anatomical conditioning, MSE/L1/SSIM losses, and classifier-free guidance for T1-to-T2 synthesis; achieved **27.47 dB PSNR** on BraTS 2020, exceeding the strongest of 6 baselines by 1.74 dB; **first-author paper** accepted at ICONIP 2026.

Motion Capture | Real-Time Dual-Camera Tracking — Oct 2024 - Apr 2025

- Extended an offline, single-camera object-pose pipeline to live **dual-camera processing** with real-time 6-DoF tracking at **60 FPS**, evaluated on 3 objects across changing scenes.
- Built a **Flask/Socket.IO and Vue.js** interface to persist and reuse object-reference data, allowing previously registered objects to be tracked without repeating video capture and reference-data preprocessing.

PUBLICATIONS

- **Lu, Z.,** Huang, M., & He, X. CMCD: Conditional Diffusion Model for Medical Image Modality Translation. International Conference on Neural Information Processing (ICONIP), 2026 (accepted).
- Fan, S., **Lu, Z.,** Yan, Z., & Hu, L. Interactions of Three Berberine Mid-Chain Fatty Acid Salts with Bovine Serum Albumin (BSA): Spectroscopic Analysis and Molecular Docking. International Journal of Biological Macromolecules, 274, 133370, 2024.

SKILLS

Languages: Python, C++, JavaScript, Java, C#, SQL
Software Engineering: Data structures and algorithms, OOP, Flask, FastAPI, Vue.js, Socket.IO, Git, Linux
ML & GenAI: PyTorch, OpenCV, LLM agents, diffusion models, Hugging Face, OpenAI API, Transformer